

ENVELOPE FOLLOWER

The envelope follower generates a corresponding "control voltage curve" from the envelope of any given audio signal, which can be used to modulate any of the FORMANT's modulation inputs. Additionally, whenever the envelope reaches an adjustable threshold, a gate pulse is emitted to trigger the ADSR envelope generators.

There is often a desire to influence and control the synthesizer using external signal sources (microphone, electric guitar, electric organ, etc.). Although the FORMANT is fully voltage-controlled, external signal sources can—at best, and following prior adaptation—be fed into the audio signal path via the ES inputs.

The envelope follower converts the volume contour of an external signal source into a control voltage proportional to the input signal's volume. Additionally, a gate signal is generated from the control voltage whenever an adjustable "minimum level" (threshold) is exceeded. The gate signal available at the trigger output is intended for triggering the FORMANT ADSRs or ADSR controllers.

The Circuit.

Figure 1 shows the basic layout of the envelope follower. The circuit (Figure 3) is essentially identical to the "envelope detector" found in the ELEKTOR ring modulator. Nevertheless, the circuit's operating principle is briefly explained here.

The envelope detector (A1–A3) produces an output signal corresponding to the envelope of the input signal. The input stage is designed to be sensitive enough to process a microphone signal directly; a signal of just 10 mV is sufficient for full-scale operation. This level of sensitivity is adequate for most microphone and electric instrument outputs. Op-amp A1 boosts the input signal to the required level, with sensitivity adjustable via potentiometer P1. Following the amplifier stage, any excessively high voltages are limited by diodes D1 and D2.

J-FET op-amp A2, together with diode D3 and capacitor C4, forms a peak rectifier. Combined with the subsequent low-pass filter—which has a cutoff frequency of approximately 10 Hz—this generates a signal that precisely tracks the envelope of the input signal.

Op-amp A3 determines the level of the envelope. The voltage at the "Envelope Output" ranges from 0 V to +10 V.

Op-amp A4 is configured as a comparator and serves to generate the FORMANT-compatible gate signal.

The "Envelope" signal is applied to the non-inverting input of A4 (pin 3 of IC1). The inverting input (pin 2) is connected to the wiper of P2, which forms a voltage divider with R16; P2 is used

to set the "trigger threshold." A voltage divider is also connected to the comparator's output (pin 1); its variable component, P3, allows the gate signal amplitude to be adjusted to approximately 4.7–5.0 V. Diode D5 clamps the comparator's negative output voltage to ground potential. LED D4 provides a visual indication of the gate pulses. Whenever the envelope signal exceeds the level of the adjustable trigger threshold, the gate signal jumps from approximately 0 V to approximately 5 V.

Construction

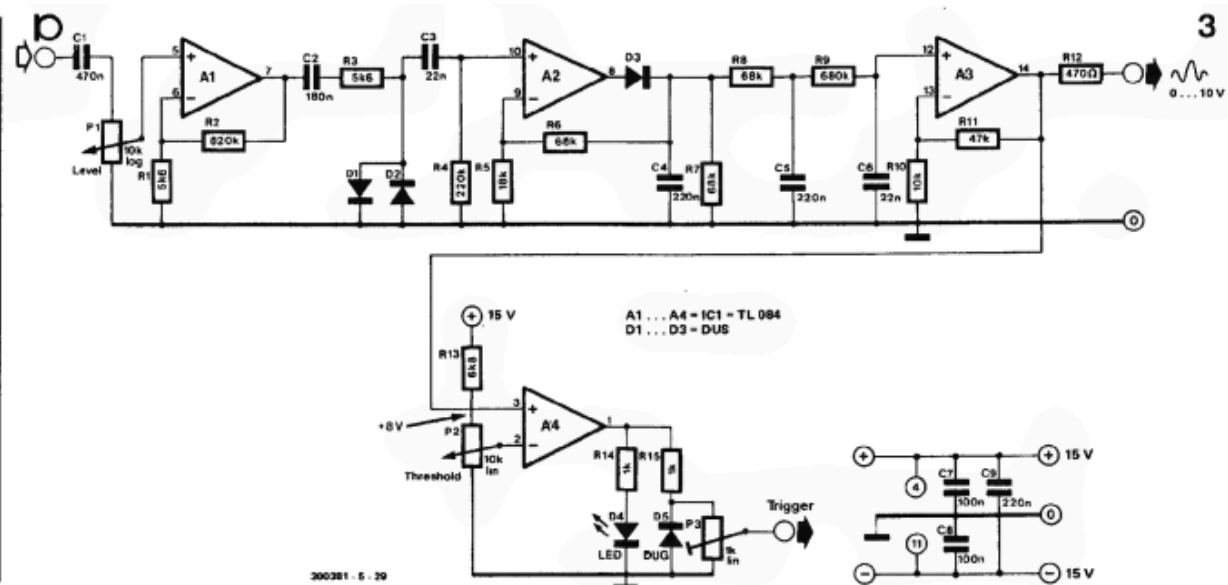
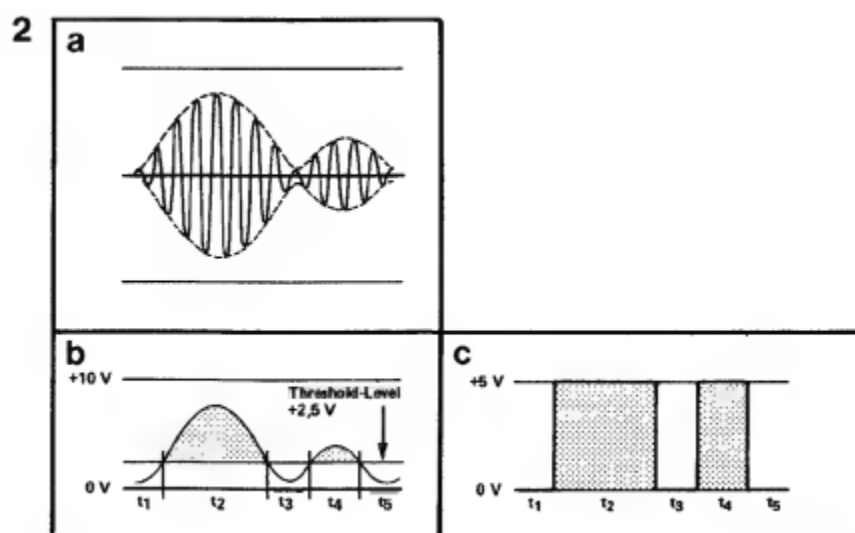
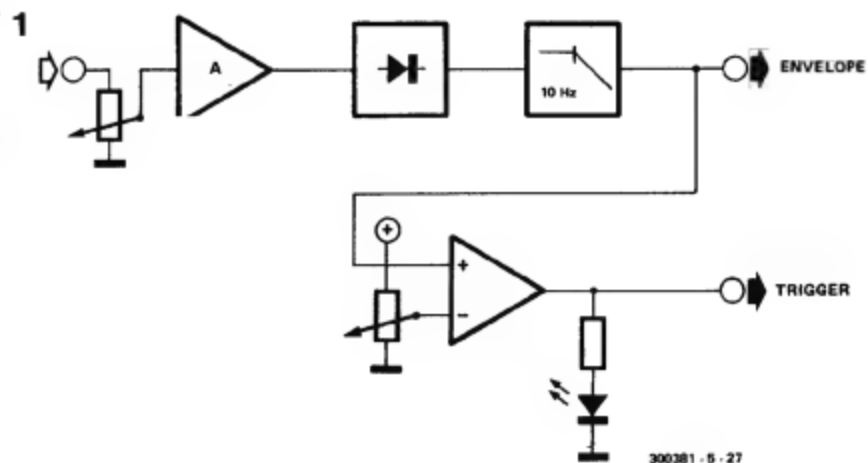
The circuit can be assembled using the Euro-format PCB (Figure 4). The J-FET IC must be handled with appropriate care and caution; using an IC socket is recommended. A front panel (Figure 5) completes the envelope follower module. As is standard for the FORMANT system, the circuit is powered by ± 15 V supply voltages and draws only a few tens of milliamperes. Consequently, the envelope follower module can be added to any existing FORMANT setup without concern.

Applications

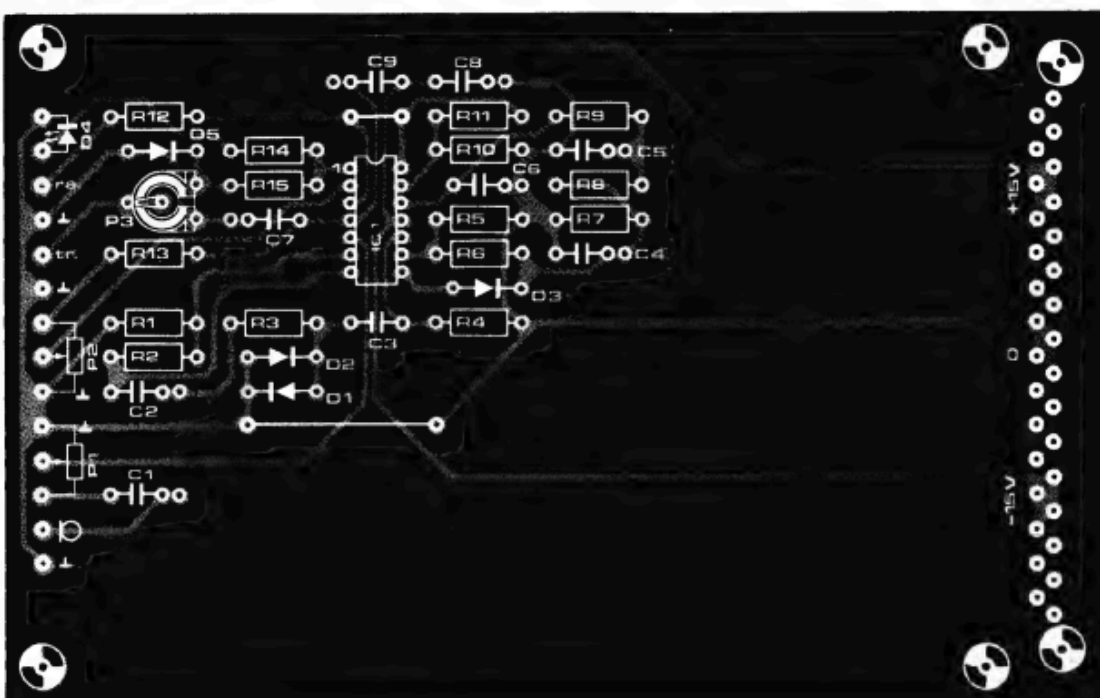
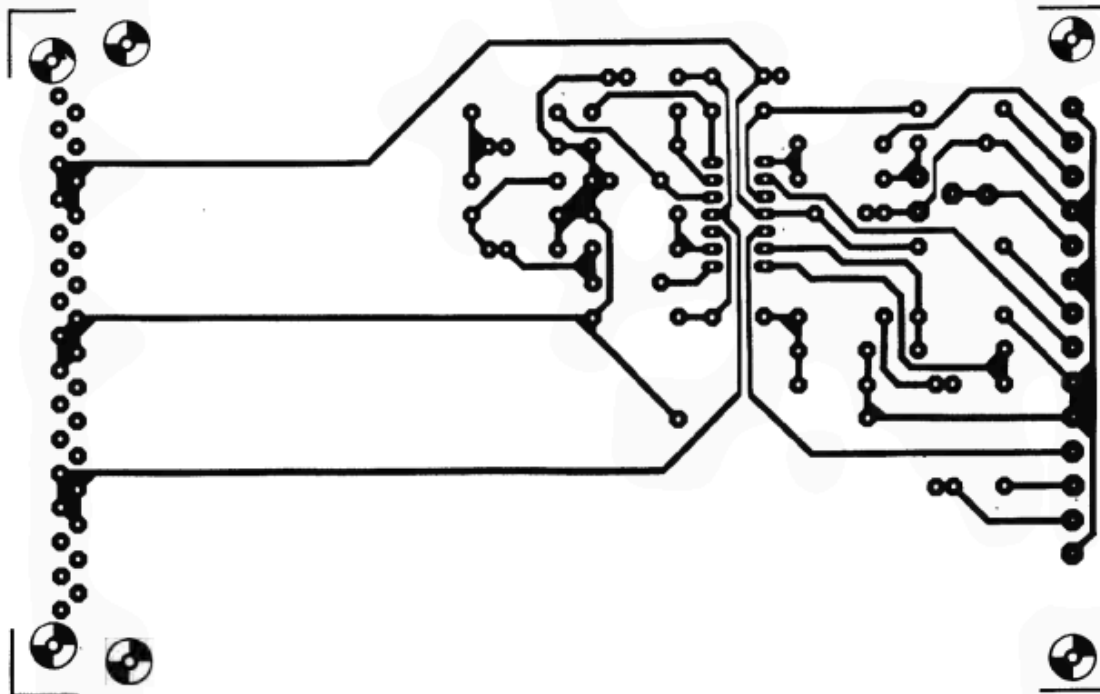
Figure 6 shows the basic circuit of the envelope follower. The potential applications are diverse. For instance, drums picked up by a microphone can control volume, pitch, or harmonic content; each drum beat can trigger an ADSR envelope. Alternatively, the FORMANT can automatically adjust its volume to match other instruments being played alongside it via the envelope follower. This can be easily achieved by using the "envelope" signal to amplitude-modulate the VCA. Furthermore, the envelope follower module can drive all other modulation and control inputs. This allows, for example, for "whistle control" of the FORMANT using the human voice or by whistling. Some ideas for building guitar and drum synthesizers are presented in the description of the MIXER (elsewhere in this chapter).

Figures

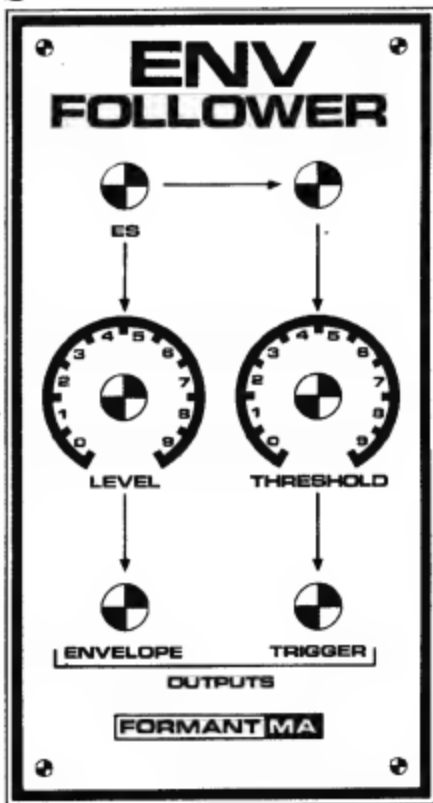
- Figure 1. Block diagram of the FORMANT envelope follower.
- Figure 2. The individual waveforms and signals illustrate the circuit's operation:
 - a) Input signal
 - b) Output signal (envelope)
 - c) Output signal (trigger)
- Figure 3. Envelope follower circuit.
- Figure 4. Layout, component placement plan, and parts list for the circuit shown in Figure 2.
- Figure 5. Proposed front panel design for the ENV-F module.
- Figure 6. Basic circuit of the envelope follower module: The envelope and trigger signals derived from the external signal control the volume or trigger ADSR contours (optionally via the ADSR controller module).



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Envelope Follower Bill of Materials

Resistors (carbon film, 5%)

R1, R3 = 5.6 k Ω

R2 = 820 k Ω

R4 = 220 k Ω

R5 = 18 k Ω

R6, R7, R8 = 68 k Ω

R9 = 680 k Ω

R10 = 10 k Ω

R11 = 47 k Ω

R12 = 470 Ω

R13 = 6.8 k Ω

R14, R15 = 1 k Ω

Potentiometers:

P1 = 10 k Ω log.

P2 = 10 k Ω lin.

P3 = 1 k Ω (trimmer)

Capacitors (film capacitors, MKH, MKS):

C1 = 470 nF

C2 = 180 nF (150 nF + 33 nF)

C3, C6 = 22 nF

C4, C5, C9 = 220 nF

C7, C8 = 100 nF

Semiconductors:

D1, D2, D3 = DUS (signal diode)

D4 = LED

D5 = DUG (general-purpose diode)

A1... A4 = IC1 = TL 084 CN (TL 074 CN)

Miscellaneous:

2 x 3.5 mm jack sockets

1 x 6.35 mm jack socket

1 x 37-pin header strip or solder pins